Circles

Class 2

Note for the teacher.

- **Aims.** Students locate the centre of a circle from two chords. They explain why the centres of all circles through two points lie on the perpendicular bisector of the segment, and why exactly one circle passes through three non-collinear points. They construct circumcircles and predict where the circumcentre lies. They explain why equal chords are equidistant from the centre, and the converse, and why a longer chord is nearer the centre.
- **NCERT links.** Theorem 5 is NCERT Theorem 1. Theorems 6, 7 and 8 keep their NCERT numbers. The questions include NCERT Exercise Sets 5.1, 5.4 and 5.5, the two *Think, Draw and Infer* questions, and End-of-Chapter Questions 12, 13, 17 and 23.
- **Prior work.** Class 1: Theorems 1 to 4, and the fact that the perpendicular bisector of every chord passes through the centre. Students use $\left(\frac{c}{2}\right)^2 + d^2 = r^2$ with confidence.
- **Materials.** Two A4 sheets, a bangle (or any round lid) and the geometry box for each student.
- **Method.** As in Class 1, students solve each *Solve first* set before the theorem is stated.

1 Finding the centre of a circle

Jamuna has drawn a circle by tracing around a bangle. She wants to find its centre. Her friend Amina says, "Fold the paper twice." Solve P1 to see why this works.

Solve first.

P1. Trace around a bangle on an A4 sheet. Draw any chord AB that is not a diameter. Hold the sheet against a window and fold it so that A falls exactly on B. Open the sheet.

(a) The crease is the perpendicular bisector of AB. Which special point of the circle must it pass through? Use a fact from Class 1.

(b) Draw a second chord CD, not parallel to AB, and fold again so that C falls on D. Where do the two creases meet?

(c) Why is one crease not enough to find the centre?

(d) What goes wrong if CD is parallel to AB?

Answers.

P1.

(a) The centre. The perpendicular bisector of every chord passes through the centre (Class 1).

(b) Both creases pass through the centre, and two lines that are not parallel meet in exactly one point. So the creases meet at the centre.

(c) One crease is a line. It contains the centre, but it contains infinitely many other points too.

(d) Both creases are perpendicular to the same direction and both pass through the centre. So they are the same line, and no single point is fixed.

FIG

Note for the teacher. Students may cut the circle out if scissors are available; folding the full A4 sheet against a window works equally well. Each crease is a diameter, and every diameter is a line of reflection symmetry of the circle. A circle also looks the same after any rotation about its centre: it has complete rotational symmetry.

Things to Remember

To find the centre of a circle, draw two chords that are not parallel. Their perpendicular bisectors meet at the centre.

2 Circles through two points

Solve first.

P2. Points A and B are 8 cm apart, and M is the midpoint of AB. Draw a rough diagram for each part.

(a) Find the centre and the radius of the smallest circle that passes through A and B.

(b) A circle of radius 5 cm passes through A and B, and its centre is O. Explain why O lies on the perpendicular bisector of AB. Find OM.

(c) How many circles of radius 5 cm pass through A and B?

(d) Can a circle of radius 3 cm pass through A and B? Give a reason.

P3. Use P2 to answer these questions.

(a) Where do the centres of all the circles through A and B lie?

(b) As the centre moves away from M, does the radius increase or decrease?

(c) Is there a largest circle through A and B?

Answers.

P2.

(a) The centre is M and the radius is 4 cm. Then AB is a diameter. A smaller radius is impossible: AB is a chord, and no chord is longer than the diameter.

FIG

(b) $OA = OB$ (radii), so O is equidistant from A and B. Such a point lies on the perpendicular bisector of AB. In right $\triangle OMA$, $OM^2 = 5^2 - 4^2 = 9$, so $OM = 3$ cm.

(c) Two: one centre 3 cm on each side of AB, on the perpendicular bisector.

(d) No. The diameter would be 6 cm, but the chord AB would be 8 cm.

P3.

(a) On the perpendicular bisector of AB. Every point on it is equidistant from A and B, so every such point is the centre of a circle through A and B.

(b) The radius increases: $r^2 = OM^2 + 4^2$ grows as OM grows.

(c) No. The centre can move as far as we like, so the radius has no largest value. The circles become flatter near AB.

Things to Remember

Infinitely many circles pass through two points A and B. Their centres lie on the perpendicular bisector of AB. The smallest has AB as a diameter.

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3 Circles through three points

Solve first.

P4. A, B and C lie on a straight line, with $AB = 4$ cm and $BC = 6$ cm.

(a) Draw the perpendicular bisectors of AB and BC.

(b) Do they meet? Give a reason.

(c) Can a circle pass through A, B and C?

P5. Draw any triangle ABC on an A4 sheet.

(a) Construct the perpendicular bisectors of AB and BC. Mark their meeting point O.

(b) Measure OA, OB and OC.

(c) Explain, without measuring, why $OA = OB = OC$.

(d) Does the perpendicular bisector of CA also pass through O? Why?

Answers.

P4.

(b) No. Both bisectors are perpendicular to the same line ABC, so they are parallel.

FIG

(c) No. A centre would have to lie on both bisectors, and parallel lines have no common point.

P5.

(b) The three lengths are equal (within about 1 mm).

(c) O is on the perpendicular bisector of AB, so $OA = OB$. O is on the perpendicular bisector of BC, so $OB = OC$. Hence $OA = OB = OC$.

(d) Yes. $OA = OC$, so O is equidistant from C and A and lies on the perpendicular bisector of CA.

Note for the teacher. P5(d) shows that the three perpendicular bisectors of a triangle always meet at one point. Students often draw all three and treat the third as a check of accuracy only.

Theorem 5 (NCERT Theorem 1). *There is a unique circle passing through three non-collinear points.*

FIG

Why it works.

Given: three points A, B and C that are not on one line.
To show: exactly one circle passes through A, B and C.

1. The centre of any circle through A and B lies on the perpendicular bisector of AB (P3). The centre of any circle through B and C lies on the perpendicular bisector of BC.
2. AB and BC are not on one line, so their perpendicular bisectors are not parallel. They meet in exactly one point O.
3. $OA = OB = OC$, so the circle with centre O and radius OA passes through all three points.
4. Any circle through the three points must have its centre at this point O and radius OA. So there is only one such circle.

This circle is the **circumcircle** of $\triangle ABC$. Its centre O is the **circumcentre**, and its radius is the **circumradius**.

Solve first.

P6. Construct each triangle and its circumcircle. Say whether the circumcentre lies inside the triangle, on a side, or outside the triangle.

(a) $AB = 5$ cm, $\angle A = 70°$, $\angle B = 60°$.
(b) $PQ = 6$ cm, $\angle P = 90°$, $PR = 8$ cm.
(c) $XY = 5$ cm, $\angle X = 100°$, $XZ = 4$ cm.

Answers.

P6.

(a) Acute-angled triangle ($\angle C = 50°$): the circumcentre is inside. Circumradius about 3.3 cm.
(b) Right-angled triangle: the circumcentre is the midpoint of the hypotenuse QR. $QR = 10$ cm, so the circumradius is 5 cm.
(c) Obtuse-angled triangle: the circumcentre is outside, beyond the longest side YZ. Circumradius about 3.5 cm.

FIG

Note for the teacher. Give one triangle to each group and let the groups compare. Part (b) needs one more fact before it can be proved: the angle in a semicircle is 90°. That is done in Class 3.

Things to Remember

The circumcentre of a triangle lies inside an acute-angled triangle, at the midpoint of the hypotenuse of a right-angled triangle, and outside an obtuse-angled triangle.

Example 1. A potter finds a broken piece of a round clay plate. Explain how to find the centre and the radius of the full plate.

FIG

Given: a piece of the rim of a circle.
To find: the centre and the radius of the circle.

1. Place the piece on paper and trace its curved edge. Mark three points A, B and C on the traced arc.
2. Draw the chords AB and BC, and construct their perpendicular bisectors. They meet at O.
3. O is the centre: the perpendicular bisector of every chord passes through the centre, and only one circle passes through A, B and C (Theorem 5).
4. Measure OA. This is the radius.

Answer: the centre is the point where the two perpendicular bisectors meet, and the radius is OA.

Exercise 2A

1. Construct $\triangle ABC$ with $AB = 6$ cm, $BC = 7$ cm and $CA = 7$ cm. Construct its circumcircle, and let the circumcentre be O. Measure OA, OB and OC.

2. Points A and B are 10 cm apart. What is the least possible radius of a circle through A and B? Is there a greatest possible radius? Explain.

3. Points P and Q are 6 cm apart. A circle of radius 5 cm passes through P and Q.

   (a) Find the distance of its centre from the line PQ.

   (b) How many such circles are there?

   (c) Explain why no circle of radius 2 cm passes through P and Q.

4. A, B and C lie on a straight line. Is there a point P with $PA = PB = PC$? Explain using the perpendicular bisectors of AB and BC. Can a straight line meet a circle in three points?

5. Three villages lie at the corners of a triangle. The panchayat wants to build a water tank at the same distance from all three. Explain how to find the place on a map. What happens if the three villages lie on one straight road?

6. Kiran wants to find the centre of a circular plate drawn on paper. He draws two parallel chords and folds the sheet so that the ends of each chord meet. Find his mistake, and correct the method.

7. The circumcircle of $\triangle ABC$ is drawn. Can another triangle, congruent to $\triangle ABC$, have the same circumcircle? Explain.

8. The sides of a right-angled triangle are 5 cm, 12 cm and 13 cm. Where is its circumcentre? Find the circumradius.

9. The circumcentre of $\triangle ABC$ lies on the side BC. What kind of triangle is $\triangle ABC$? Draw a rough figure to support your answer.

Answers.

1.

$\triangle ABC$ is isosceles and acute-angled, so O lies inside it, on the axis of symmetry through C (the perpendicular bisector of AB).
Answer: $OA = OB = OC \approx 3.9$ cm.

FIG

2. The least radius is 5 cm, with AB as a diameter. There is no greatest radius: the centre can be anywhere on the perpendicular bisector of AB, as far away as we like (P3).

3.

(a) The centre O is on the perpendicular bisector of PQ, which meets PQ at its midpoint M. $PM = 3$ cm. In right $\triangle OMP$, $OM^2 = 5^2 - 3^2 = 16$, so $OM = 4$ cm.

(b) Two: one centre on each side of PQ.

(c) Its diameter would be 4 cm, but $PQ = 6$ cm would be a chord, and no chord is longer than the diameter.

FIG

Answer: (a) 4 cm; (b) two.

4. No. P would lie on the perpendicular bisectors of AB and of BC. Both are perpendicular to the line ABC, so they are parallel and never meet. So no circle passes through three collinear points, and a line meets a circle in at most two points.

5. Join the villages to form a triangle and construct the perpendicular bisectors of two sides. The tank goes at the point where they meet, the circumcentre. If the villages lie on one straight road, the bisectors are parallel, and no such place exists (Question 4).

6.

Both creases are perpendicular to the parallel chords and both pass through the centre. So they are the *same* line, and they do not fix a point. Kiran must draw two chords that are *not* parallel.

FIG

7. Yes, infinitely many. Turn $\triangle ABC$ about the circumcentre O through any angle. The image is congruent to $\triangle ABC$, and its vertices stay on the same circle. Reflecting in a diameter also works.

8.

$5^2 + 12^2 = 169 = 13^2$, so the triangle is right-angled, and 13 cm is the hypotenuse. The circumcentre is the midpoint of the hypotenuse (P6).
Answer: circumradius $= \dfrac{13}{2} = 6.5$ cm.

FIG

9. A right-angled triangle, with the right angle at A. In P6 the circumcentre was on a side only for the right-angled triangle, and then BC is the hypotenuse.

Note for the teacher. Question 9 is checked by construction here. Students prove it in Class 3, using the angle in a semicircle.

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4 Chords and their distance from the centre

Recall from Class 1: the distance of a chord from the centre is the length of the perpendicular from the centre to the chord.

Solve first.

P7. A circle has centre O and radius 5 cm. Chords AB and CD are each 8 cm long. $OM \perp AB$ and $ON \perp CD$.

(a) Find OM and ON.

(b) What do you notice?

(c) Would the answer to (b) change for two other equal chords of this circle? Why?

P8. In the same circle, chords PQ and RS are each 4 cm from O. Find PQ and RS.

P9. In the same circle, copy and complete the table.

Distance from O (cm)	0	3	4	5
Length of chord (cm)				

(a) As the chord moves away from the centre, what happens to its length?

(b) Which chord is the longest? Where does it lie?

(c) Of two unequal chords, which one is nearer the centre?

Answers.

P7.

(a) $AM = 4$ cm (the perpendicular from the centre bisects the chord, Theorem 4). In right $\triangle OMA$, $OM^2 = 5^2 - 4^2 = 9$, so $OM = 3$ cm. In the same way, $ON = 3$ cm.

FIG

(b) The equal chords are at equal distances from O.

(c) No. For any chord of length 8 cm the calculation uses the same radius and the same half chord, so the distance is always 3 cm.

P8. In right $\triangle OMP$, $PM^2 = 5^2 - 4^2 = 9$, so $PM = 3$ cm and $PQ = 6$ cm. In the same way, $RS = 6$ cm. Chords at equal distances from the centre are equal.

P9.

Distance from O (cm)	0	3	4	5
Length of chord (cm)	10	8	6	0

(a) It becomes shorter. At distance 5 cm the "chord" is a single point.

(b) The diameter, 10 cm. It passes through the centre (distance 0).

(c) The longer chord.

Theorem 6 (NCERT Theorem 6). *Chords of a circle having the same length are all at the same distance from the centre of the circle.*

FIG

Why it works.

Given: chords AB and CD of a circle with centre O, $AB = CD$, $OM \perp AB$ and $ON \perp CD$.
To show: $OM = ON$.

1. M and N are the midpoints of AB and CD (Theorem 4). Since $AB = CD$, $AM = CN$.
2. In $\triangle OMA$ and $\triangle ONC$: $\angle OMA = \angle ONC = 90°$, $OA = OC$ (radii) and $AM = CN$.
3. So $\triangle OMA \cong \triangle ONC$ (RHS), and $OM = ON$.

Theorem 7 (NCERT Theorem 7). *Chords of a circle that are equidistant from the centre have equal length.*

Why it works.

Given: chords AB and CD of a circle with centre O, $OM \perp AB$, $ON \perp CD$ and $OM = ON$ (same figure).
To show: $AB = CD$.

1. In $\triangle OMA$ and $\triangle ONC$: $\angle OMA = \angle ONC = 90°$, $OA = OC$ (radii) and $OM = ON$ (given).
2. So $\triangle OMA \cong \triangle ONC$ (RHS), and $AM = CN$.
3. $AB = 2AM$ and $CD = 2CN$ (Theorem 4), so $AB = CD$.

Theorem 8 (NCERT Theorem 8). *Let AB and CD be two chords of a circle with centre O, with $AB > CD$. Then the distance of AB from O is less than the distance of CD from O.*

Why it works.

Given: $AB > CD$, $OM \perp AB$ and $ON \perp CD$ (same figure, with AB longer).
To show: $OM < ON$.

1. By Pythagoras theorem, $OM^2 + AM^2 = OA^2$ and $ON^2 + CN^2 = OC^2$. Since $OA = OC$, $OM^2 + AM^2 = ON^2 + CN^2$.
2. $AM = \dfrac{AB}{2}$ and $CN = \dfrac{CD}{2}$. Since $AB > CD$, $AM > CN$, so $AM^2 > CN^2$.
3. The two sums are equal, so $OM^2 < ON^2$, and $OM < ON$.

Things to Remember

For a chord at distance d from the centre of a circle of radius r, the length of the chord is $2\sqrt{r^2 - d^2}$. Equal chords are at equal distances from the centre, and the longer of two chords is nearer the centre. The diameter is the longest chord.

Example 2. A circle has centre O and radius 17 cm. Chord AB is 30 cm long and chord CD is 16 cm long. Which chord is nearer the centre? Find the distance of each chord from O.

FIG

Given: $r = 17$ cm, $AB = 30$ cm, $CD = 16$ cm; $OM \perp AB$, $ON \perp CD$.
To find: which chord is nearer O; OM and ON.

1. $AB > CD$, so AB is nearer the centre (Theorem 8).
2. $AM = 15$ cm (Theorem 4). In right $\triangle OMA$, $OM^2 = 17^2 - 15^2 = 289 - 225 = 64$, so $OM = 8$ cm.
3. $CN = 8$ cm. In right $\triangle ONC$, $ON^2 = 17^2 - 8^2 = 289 - 64 = 225$, so $ON = 15$ cm.

Answer: AB is nearer the centre; it is 8 cm from O, and CD is 15 cm from O.

Example 3. Chords AB and CD of a circle with centre O are each 6 cm from O, and $AB = 16$ cm. Find CD and the radius of the circle.

FIG

Given: $OM \perp AB$, $ON \perp CD$, $OM = ON = 6$ cm, $AB = 16$ cm.
To find: CD and the radius OA.

1. The chords are equidistant from the centre, so $CD = AB = 16$ cm (Theorem 7).
2. $AM = 8$ cm (Theorem 4). In right $\triangle OMA$, $OA^2 = 6^2 + 8^2 = 100$, so $OA = 10$ cm.

Answer: $CD = 16$ cm and the radius is 10 cm.

Draw a rough figure for every question in Exercise 2B and mark all the given measurements on it.

Exercise 2B

1. A circle has centre O and radius 10 cm. Chords AB and CD are each 12 cm long. Find the distance of AB from O. Which theorem gives the distance of CD without a second calculation?
2. Chords PQ and RS of a circle with centre O are each 5 cm from O, and $PQ = 24$ cm. Find RS and the radius.
3. Use Pythagoras theorem to explain why Theorem 6 is true. Then use it to explain why Theorem 7 is true.
4. A circle has radius 13 cm. It has chords of lengths 10 cm, 24 cm and 26 cm. Without calculating, arrange them from the nearest to the farthest from the centre. Then find each distance to check.
5. In a circle of radius 5 cm, a chord of length 8 cm is 3 cm from the centre. Priya says, "A chord half as long, 4 cm, must be twice as far from the centre, 6 cm." Find her mistake.
6. A chord of a circle of radius r is at distance d from the centre. Explain why the length of the chord is $2\sqrt{r^2 - d^2}$.
7. A circular park has radius 17 m and a fountain at its centre. Two straight footpaths cross the park, each 30 m long from edge to edge.
   (a) How far is each footpath from the fountain?

(b) The two paths are parallel and on opposite sides of the fountain. Find the distance between them.

8. Draw a circle in which a chord of length 6 cm is at a distance of 3 cm from the centre. Write your steps, and measure the radius.

9. True or false? Give a reason for each.

   (a) Two chords at the same distance from the centres of two different circles are equal.

   (b) Of two chords of a circle, the longer one is nearer the centre.

   (c) The chord farthest from the centre is the diameter.

10. In a circle with centre O, chords AB and AC are equal. Explain why the centre O lies on the bisector of $\angle BAC$.

Note for the teacher. Questions 1 to 4 give the essential practice with Theorems 6 to 8. Question 5 must be discussed in class. If time is short, set Questions 7, 8 and 10 as homework.

Answers.

1.

Given: $r = 10$ cm, $AB = CD = 12$ cm. **To find:** the distances of AB and CD from O.

FIG

1. $OM \perp AB$, so $AM = 6$ cm (Theorem 4).
2. In right $\triangle OMA$, $OM^2 = 10^2 - 6^2 = 64$, so $OM = 8$ cm.
3. $CD = AB$, so CD is also 8 cm from O (Theorem 6).

Answer: 8 cm each; Theorem 6.

2.

FIG

1. $RS = PQ = 24$ cm: the chords are equidistant from the centre (Theorem 7).
2. $PM = 12$ cm. In right $\triangle OMP$, $OP^2 = 5^2 + 12^2 = 169$, so $OP = 13$ cm.

Answer: $RS = 24$ cm, radius 13 cm.

3. Use the figure of Theorem 6, with $OM \perp AB$ and $ON \perp CD$.

1. In right $\triangle OMA$, $OM^2 = OA^2 - AM^2$. In right $\triangle ONC$, $ON^2 = OC^2 - CN^2$.
2. Theorem 6: if $AB = CD$, then $AM = CN$. Also $OA = OC$. So $OM^2 = ON^2$, and $OM = ON$.
3. Theorem 7: if $OM = ON$, then $AM^2 = OA^2 - OM^2 = OC^2 - ON^2 = CN^2$. So $AM = CN$, and $AB = 2AM = 2CN = CD$.

4.

The longest chord is nearest the centre (Theorem 8). Order: 26 cm, 24 cm, 10 cm.

FIG

1. 26 cm is the diameter: distance 0.
2. 24 cm: $d^2 = 13^2 - 12^2 = 25$, so $d = 5$ cm.
3. 10 cm: $d^2 = 13^2 - 5^2 = 144$, so $d = 12$ cm.

Answer: 0 cm, 5 cm, 12 cm.

5.

For the 4 cm chord, half the chord is 2 cm, so $d^2 = 5^2 - 2^2 = 21$ and $d = \sqrt{21} \approx 4.6$ cm, not 6 cm. In fact, no chord can be 6 cm from the centre: 6 cm is more than the radius.
Distance and length are not in proportion. They are linked by $\left(\frac{c}{2}\right)^2 + d^2 = r^2$.

FIG

6. Let M be the foot of the perpendicular from the centre O to the chord AB. Then M is the midpoint of AB (Theorem 4), and $\triangle OMA$ is right-angled at M. So $AM^2 = OA^2 - OM^2 = r^2 - d^2$, and $AM = \sqrt{r^2 - d^2}$. The chord is $AB = 2AM = 2\sqrt{r^2 - d^2}$.

7.

(a) Each path is a chord of length 30 m, so half of it is 15 m. $d^2 = 17^2 - 15^2 = 64$, so $d = 8$ m. Both paths are equal, so both are 8 m from the fountain (Theorem 6).

(b) The perpendiculars from the centre to the two parallel paths lie on one line. The paths are on opposite sides, so the distance is $8 + 8 = 16$ m.

FIG

Answer: (a) 8 m each; (b) 16 m.

8.

1. Draw $AB = 6$ cm and its perpendicular bisector, meeting AB at M.
2. Mark O on the bisector with $OM = 3$ cm.
3. Draw the circle with centre O and radius OA.

FIG

$OA^2 = 3^2 + 3^2 = 18$, so the radius is about 4.2 cm.

9.

(a) False. Theorems 6 and 7 are about one circle. Two chords 8 cm from the centres of circles of radii 10 cm and 17 cm have lengths 12 cm and 30 cm.

(b) True (Theorem 8).

(c) False. The diameter passes through the centre, so it is the *nearest* chord, at distance 0.

10.

Given: $AB = AC$, chords of a circle with centre O. **To show:** $\angle OAB = \angle OAC$.

FIG

1. In $\triangle OAB$ and $\triangle OAC$: $OB = OC$ (radii), $AB = AC$ (given) and OA is common.
2. So $\triangle OAB \cong \triangle OAC$ (SSS), and $\angle OAB = \angle OAC$.
3. So AO bisects $\angle BAC$, and O lies on the bisector.

(Another way: $AB = AC$, so the chords are equidistant from O (Theorem 6). A point equidistant from the two arms of an angle lies on its bisector.)

5 Further questions

F1. Find the length of a chord of a circle of radius 7 cm that is 6 cm from the centre. Leave the answer in square-root form.

F2. Two equal chords AB and CD of a circle intersect at P. Show that $AP = CP$ and $PB = PD$.

F3. P is a point inside a circle with centre O (P is not O). Show that, of all the chords through P, the shortest one is perpendicular to OP.

Answers.

F1. Half the chord is $\sqrt{7^2 - 6^2} = \sqrt{13}$ cm, so the chord is $2\sqrt{13}$ cm (about 7.2 cm).

F2.

FIG

1. Draw $OM \perp AB$ and $ON \perp CD$. $AB = CD$, so $OM = ON$ (Theorem 6), and M, N are the midpoints (Theorem 4).
2. In $\triangle OMP$ and $\triangle ONP$: right angles at M and N, OP common, $OM = ON$. So $\triangle OMP \cong \triangle ONP$ (RHS), and $MP = NP$.
3. $AM = CN$ (halves of equal chords). With P placed as in the figure, $AP = AM + MP = CN + NP = CP$.
4. $PB = AB - AP = CD - CP = PD$.

F3.

FIG

1. Let AB be any chord through P, and let $OM \perp AB$. In right $\triangle OMP$, $OM \le OP$, with equality only when $M = P$, that is, when $AB \perp OP$.
2. A chord farther from the centre is shorter (Theorem 8). So the chord through P that is farthest from O, at distance OP, is the shortest. It is the chord perpendicular to OP.

Note for the teacher. F3 leads into Class 3, which turns from chords to the angles that arcs subtend at the centre and on the circle.
